NATIONAL UNIVERSITY



Fourth Year Syllabus

Department of Biochemistry and Molecular Biology

Four-Year B.Sc. (Honours) Course

Effective from the Session: 2013–2014

National University

Subject: Biochemistry and Molecular Biology
Syllabus for Four Year B. Sc Honours Course
Effective from the Session: 2013-2014

Year wise Papers and marks distribution.

FOURTH YEAR

Paper Code	Paper Title	Marks	Credits
242901	Cell Biology	100	4
242903	Molecular Biology-III	100	4
242905	Plant Biochemistry & Biotechnology	100	4
242907	Immunology	100	4
242909	Industrial Biotechnology	100	4
242911	Virology & Cancer Biology	100	4
242913	Biochemistry of Natural Products	100	4
242915	Biostatistics	100	4
242916	Practical	100	4
242918	Viva-voce	100	4
	Total =	1000	40

Detailed Syllabus

Paper Code	242901	Marks : 100	Credits : 4	Class Hours : 60 hrs.
Paper Title :	Cell Biology			

1. Brief history of cell theory, differentiation between Prokaryotic and eukaryotic cell str. Some important characteristics of the cell such as selectively permeable membrane. Self assembly and replication. Cell organells and compartmentalization, cell cycle. Cell membrane and its organization.

2. Cell growth and cell division : Control of cell division, tumor viruses as tools for studying the control of the cell cycle events in the S phase, the logic of the cycle, cell division.

3. The Cytoskeleton : Muscle contraction, ciliary movement, general features of microtubules and actin filaments as dynamic assemblies, microtubule organizing centers and microtuble associated proteins, actin filaments and actin binding proteins in nonmuscle cells, intermediate filaments, organization of the cytoskeleton.

4. Cell-cell adhesion : The extracellular matrix, intercellular recognition and cell adhesion cell junctions.

5. Chemical signaling between cells : Cell membrane and its organization. Cell signalling different strategies of chemical signaling, local chemical mediators, hormones and neurotransmitters, signaling mediated by intracellular receptors, mechanism of steroid hormone action. Signaling mediated by cell surface receptors, cyclic AMP and calcium ions as second messengers, involvement of G-proteins in signal transduction, target cell adaptation Electrical signals in nerve cells.

6. Germ cells and fertilization : The benefits of sex, meiosis, gametes, and fertilization.

7. Cell communication : Cascades of induction reciprocal and sequential inductive events, instructive and permissive interactions, epithelial and mesenchymal interaction. Paracrine factors inducer molecules, Cell death pathways, Cross talk between signaling .pathways.

8. Differentiated cells and the maintenance of tissues : Maintenance of the differentiated state, tissues with .permanent cells, renewal by simple duplication, renewal by stem cells, epidermis, renewal by pluripotent stem cells, blood cell formation, quiescent stem cells, skeletal muscle, soft cells and tough martix, growth turnover, repair skeletal connective tissue, territorial stability in the adult body.

9. Some of the techniques used in the study of the cell to cell interaction and intracellular interaction of the macromolecules. Stem cells its application in diseases.

Books Recommends:

1. *MOLECULAR BIOLOGY OF THE CELL*
Bruce Alberts, Dennis Bray, Julian Lewis, Martin Raff,
Keith Roberts, James D. Watson
Garland Publishing Inc.

2. *CELL AND MOLECULAR BIOLOGY*
E.D.P. DeRobertis and E.M.F. DeRobertis.
Wavertev.

3. *MOLECULAR CELL BIOLOGY*
H. Lodish S.L. Berk, P. Ziparsky, D. Matsudaira, D. Baltimore, J. Darnell
W.H. Freeman & Company.

Paper Code	242903	Marks : 100	Credits : 4	Class Hours : 60 hrs.
Paper Title :	Molecular Biology-III			

1. Gene Structure : Interrupted genes, organization of exons and introns, distribution of genes, organization of gene families, variations in individual genomes, organization of genes in the organelles. Repetitive genes, the special features of metaphase chromosome, DNA protein interaction in centromere and telomere.
2. DNA loss and amplification as a mechanism of gene regulation, DNA rearrangement and gene shuffling.
3. Mobile genetic elements : Transposons, retroposons their characteristics & function.
4. Gene expression : Tissue specific expression of proteins and messenger RNAs, post transcriptional events.
5. Regulation at the transcriptional level : Regulation of RNA splicing, RNA editing, Regulation of RNA transport, stability and translation.
6. Transcriptional control of DNA sequence elements : Short sequence elements located within or adjacent to the gene promoter. Enhancers, negative acting sequence elements locus control regions. Regulation by RNA pol I & III. DNA binding by transcription factors.
7. Transcription control of chromatin structure : Changes in DNA methylation, alteration in histones, changes in chromatin structure.
8. Recombinant DNA : Enzymes for DNA manipulation, vectors for cloning the genes, cloning of DNA fragments, polymerase chain reaction, RTPCR, identification of human disease genes.
9. Epigenetics : Epigenetics and chromatin dynamics, silencing, transcriptional landscapes and genomics, memory of transcriptional states, stem cells and reprogramming, maintenance of (Epi)genome integrity, epigenetics and Cancer.

Books Recommended :

1. *MOLECULAR BIOLOGY OF THE CELL*
   Bruce Alberts, Dennis Bray, Julian Lewis, Martin Raff,
   Keith Roberts, James D. Watson
   Garland Publishing Inc.
2. *GENS I-VII*
   Benjamin Lewin
   Oxford University Press.
3. *MOLECULAR COLONING*
   A Laboratory Manual
   J. Sambrook, E.F. Fritsch, T. Maniatis.
   Cold Spring Harbor Laboratory Press
4. *MOLECULAR BIOTECHNOLOGY*
   Principles and Applications of Recombinant DNA
   B.R. Glick and J. J. Pasternak
   A.S.M. Press.

Paper Code	242905	Marks : 100	Credits : 4	Class Hours : 60 hrs.
Paper Title :	Plant Biochemistry & Biotechnology			

1. Plant cell and metabolism specialities, Cellular and subcellular compartmentation. Plant cell and organelle structure, role of compartmentation in regulation of metabolism, unique aspects of plant metabolism and their impact on mrtabolic flux, transporters in metabolic flux.
2. Photosynthesis. Light reaction, electron transport, dark reactions, Calvin cycle and regulation, Photorespiration, C3, C4 and CAM metabolism and their comparison. Role of carbohydrate metabolism in plants in normal and stress conditions.
3. Plant respiration specialities : Cyanide insensitive respiration; Glyoxylate cycle.
4. Nitrogen and sulfur assimilation and metabolism. Role of transporters.
5. Plant hormones. Physiology and importance of auxins, cytokinins, gibberellins, abscises acid, ethylene, brassinsteroids and polyamines. Basic concepts of plant cell signaling and stress physiology. Role of jasmonic acid, salicylic acid, nitric oxide and phospholipids.
6. Secondary metabolites, alkaloids and plant phenolics. Their production and commercial application. Medicinal plants and their uses.
7. Plant cell culture and applications : Manipulation at cellular level, totipotency of plant cells, somatic embryogenesis, organogenesis, recalcitrant plants, micropropagation and applications. Disease free plants, protoplast culture and fusion with reference to cybrids and cytoplasmic male sterility, anther culture and applications for breeding, commercialization of tissue culture technology, plant tissue culture as a basis for genetic engineering.
8. DNA markers and application for breeding : Fingerprinting for assessment of germplasm, concept of polymorphism, mapping and breeding populations, linkage of marker to trait of interest, marker-aided selection for breeding.
9. Plant genetic transformation-prospects and potential : Current status, characters transformed, techniques for plant transformation such as agrobacterium-mediated and biolistics. Use of constitutive, tissue-specific and stress-specific promoters for transformation, molecular assessment of transgenic status and inheritance of transgenes, gene silencing, current status of chloroplast transformation and advantages, plants as bioreactors and vaccine production systems, biosafety issues, GM crops.
10. Discovery/cloning of plant genes : Probe-based screening, genomic and proteomic approaches, map-based cloning, transposon tagging, isolation by T-DNA insertion, functional characterization by gene mutagenesis/silencing.

Book Recommanded :

MOLECULAR BIOLOGY OF THE CELL
Bruce Alberts, Dennis Bray, Julian Lewis, Martin Raff,
Keith Roberts, James D. Watson
Garland Publishing Inc.
2. *GENS I-VII*
Benjamin Lewin
Oxford University Press.
3. *MOLECULAR COLONING*
A Laboratory Manual

J. Sambrook, E.F. Fritsch, T. Maniatis.
Cold Spring Harbor Laboratory Press

4. *MOLECULAR BIOTECHNOLOGY*
Principles and Applications of Recombinant DNA
B.R. Glick and J. J. Pasternak
A.S.M. Press.

Paper Code	242907	**Marks : 100**	**Credits : 4**	**Class Hours : 60 hrs.**
Paper Title :	**Immunology**			

1. Introduction to immunology : Adaptive and innate immunity, cells involved in the immune response, soluble mediators of immunity- complement, cytokines and antibodies; antigens, immune responses, clonal selection, mitogens, immune effector mechanisms- inflammation, chemotaxis and phagocytosis; defences against extracellular and intracellular pathogens, vaccination, immunopathology.

2. Immunoglobulins : Distribution of major human immunoglobulins, immunoglobulin classes and subclasses, physicochemical properties and functions of human immunoglobulin classes, general properties of immunoglobulins, antibody structure, antibody effector functions, structure of immunoglobulin in relation to function, enzymatic cleavage of human IgG1, structure in relation to antigen binding, genetic basis of antibody heterogenicity.

3. Antigens : Chemical basis of antigenicity, immunogenicity, antigenic determinants, haptens, intraction of antibodies with antigens, antibody affinity and avidity, antibody specificity and cross-reactivity, physiological significance of high and low affinity antibodies.

4. Complement : Activities of complement proteins, activation of complement, classical pathway, regulation of classical pathway activation, alternative pathway of complement activation, the lectin pathway, functions of C5a and C3a, formation and regulation of the membrane attack complex, regulation of the amplification loop, biological effects of complement.

5. Techniques to study antigen-antibody interactions : Precipitation, agglutination, immunodiffusion, single radial immunodiffusion, double immunodiffusion, immunoelectrophoresis, countercurrent electrophoresis, rocket electrophoresis, immunofluorescence, complement fixation, radioimmunoassay, enzyme-linked immunosorbent assay.

6. Immunogenetics : Immunoglobulin(Ig) gene structure, mechanism and regulation of Ig gene recombination (assembly) and expression, generation of the antibody diversity, Ig gene statement, class switching.

7. Vaccination : Active and passive immunization: live, attenuated vaccines; inactivated or killed vaccines; conjugate vaccines; DNA vaccines; recombinant vector vaccines; primeboost strategies.

8. Autoimmunity and autoimmune disease : pathogenesis of autoimmunity: genetic susceptibility to autoimmunity; role of infections in autoimmunity, pathogenesis, control mechanisms, treatment; effector cell assays.

9. Allergy and Hypersensitivity : Coombs and Gell classification.

Type I -Immediate hypersensitivity : induction and effector mechanisms in type I hypersensitivity, allergens; atopy, IgE in type I hypersensitivity; control of lgE production; role

of T cells in the immune responses to inhalant allergen; role of mast cells, basophils and eosinophils in type I hypersensitivity; mast cell triggering, mediators derived from mast cells, reactions of immediate hypersensitivity, genetic susceptibility to immediate hypersensitivity, , immunopathology of allergic disorders, factors involved in the development of allergy; diagnosis: skin test; immunotherapy and new approaches for treatment.

Type II- Antibody dependent cytotoxicity : mechanism of damage, reactions against blood cells and platelets, haemolytic disease of the newborn, autoimmune haemolytic anaemia, hyper acute graft rejection; treatment.

Type III-Immune complex mediated hypersensitivity : types of immune-complex disease; inflammatory mechanisms in type III hypersensitivity; experimental models of immune-complex disease; persistence of immune-complexes; deposition of complexes in tissues; detection of immune complexes.

Type IV-Delayed hypersensitivity : contact hypersensitivity; tuberculin-type hypersensitivity; granulomatous hypersensitivity; cellular reactions in delayed hypersensitivity, diseases manifesting delayed hypersensitivity.

10. Tumor Immunology : Immune surveillance, tumor antigens, tumor associated antigens, immune responses to tumors, immune evasion by tumors, immunodiagnosis, immunotherapy for tumors.

Books Recommended :

1. *Immunology : An Introduction*
   L.R.Tizard,
   Saunders College Pub.
2. *Immunology*
   L.E. Hood, L. Weissman, W.B. Wood, J.H. Wilson
   Benjamin/Cummings
3. *Text Book of Immunology*
   J.T. Barrett
   The CV Mosby Company
4. *Immunology*
   Ivan M. Roitt, J. Brostoff and D.K. Male
   Mosby, Harcourt Publishers
5. *Basic and Clinical Immunology*
   D.P. Stites, J.D. Stobo and J.V. Wells.
   Lange Medical Pub.
6. *FundamentalsoOf Immunology*
   O.G. Bier, W.D. De Silva, D. Gotze and I. Mota.
   Springer Verlag
7. *Practical Immunology, 3rd Edition*
   L. Hudson & F.C. Hay

Paper Code	242909	**Marks : 100**	**Credits : 4**	**Class Hours : 60 hrs.**
Paper Title :	**Industrial Biotechnology**			

1. Fermentation technology : Principles of Microbial growth, with merits & demerits different types of fermentation; genetic enrichment of microbes for industrial use; different bioreactor

designs and their operations, inoculums preparation: mode of fermentation; function process with specific control system; harvesting microbial cells, disruptin microbial cells, downstream processing or product extraction fermentation technology for the production of single cell protein (SCP); biofuel: ethanol, methane.

2. Immobilized biocatalysts : Different types biocatalysts, different immobilization processes of biocatalysts: adsorption, covalent binding, entrapment, encapsulation and covalent binding, applications of immobilized biocatalysts, multienzyme systems.

3. Bioremediation and biomass utilization : Agro-industrial waste recycling, hydrocarbon and aromatic transformations, microbial degradation of xenobiotics. genetic engineering of bio-degradative. pathways.

4. Animal cell-culture : Animal cell culture, characteristics, culture design and significance, application in monoclonal antibody production; hybridoma cell lines, expression of engineered proteins in cell culture, application in introduction of a particular trait into an animal: embryo tranfer technology, invitro fertilization (IVE) technology, embryo cloning.

5. Micerobial Toxins and Insecticides : Insecticidal toxin of *B.thuringiensis,* mode of action and use, enginnering of *B.thuringiensis,* toxin gene, baculoviruses as biocontrol agents.

6. Traditional Application of Food Biotechnology : Fermented foods: e.g. dairy products, oriental fermentations, alcocholic beverages, food ingredients.

7. Food Processing and Control : Food preservation by heating, chilling, freezing, dehydration and ionising radiation, Packaging materials. Processing of selected food products, Food laws and standards, Concepts of food qulity and saflety. Factors which affect quality and safety, Food hazards, Process control, application of principles of food hygiene and relevant codes of practice/guidelines to ensure quality and safety. Product development and sensory perception. Food Flavors, additives and supplements.

Books Recommended :

1. *Principles of fermentation technology*
   Peter F Dtanbury adn Allan Whitaker;
   Pergamon press
2. *Industrial Microbiology*
   L.E. Casida JR.
   John Wiley & Sons.
3. *Industrial Biotechnology*
   B.M. Miller & W. Litsky
   McGraw-Hill Book Co.
4. *Industrial Microbiology and the Advent of Genetic Engineering*
   A. Scientific American Book.
   W.H. Freeman & Company
5. *Industrial Microbiology*
   S.C. Prescott & C.G. Dunn
   McGraw-Hill Book Co.

Paper Code	242911	**Marks : 100**	**Credits : 4**	**Class Hours : 60 hrs.**
Paper Title :	**Virology and Cancer Biology**			

1. Major groups of viruses : Bacterial, plant and animal viruses with their nomenclature and

classification.

2. Virus cultivation : Detection and Genetics: Cultivation of virus: Cell culture,, embryonated eggs, laboratory animals. Dection of virus in hosts: Measurement of Infectious units, measurement of virus particles and their components. Serological and molecular detection, plaque assay (PFU), infectious center assay, one-hit kinetics and two-hit kinetics of virus cultivation. Genetic Analysis of Virus: Classical genetic methods of engineering mutations into viruses, engineering viral genomes: viral vectors.

3. Host virus interaction : Attachment, entry and uncoating, replication, assembly and maturation, exit of virus from host cells: Mechanism of viral interation with cell. Mechanism of virus entry into cell.

4. Animal virus : Classification based on gene expression, studies on virion structure, infectivity, mode of gene expression and virus assembly of representative member of each class – herpes virus, papovavirus, hepatitis virus (HBV and HCV), picornavirus, vesicular stomatitis virus (VSV), rabies virus, reovirus, retrovirus (HIV), white spot syndrome virus (WSSV) of shrimp., Bird flue, surneflue SARS, ROTA virus.

5. Effect of animal viruses on host cells : Cytolytic effects, morphological and biochemical observations, inhibition of proteins, RNA and DNA synthesis, pattern of viral infection- acute, chronic, persistent and latent viral infection.

6. Plant virus : Structure genomic organization and molecular aspects of tobacco mosaic virus (TMV), cotton leaf curl Gemini virus (CLCuV) and potato virus Y.

7. Prevention and control of viral infection : General prevention strategies.Immunization with vaccines and antiviral drugs, mechanisms of action and limitations of use of these drugs. Interferon and its modification.

8. Introduction and overview of cancer : oncogenes and proto-oncogenes, tumor suppressor genes and hereditary cancer, basic mechanisms of cell cycle regulation, targeted ubiquitination, mammalian cell, cycle Regulation, cyclins and cyclin-dependent kinases, inhibitors of cyclin-dependent kinases.

9. The molecular biology of cancer : The retinoblastoma tumor suppressor gene, regulation of E2F transcription factors, transcription regulation by RB/E2F, The p^{53} tumor suppressor gene, regulation of p^{53} response, DNA damage and cell cycle response, apoptosis (programmed cell death), DNA tumor viruses, growth factors and receptors, non-receptor tyrosine kinases, ras signalling and adapter proteins, cancer regression by senescence

10. Cancer an epigenetic disease : DNA methylation in cancer, gene silencing and cancer, methyl CpG binding proteins and cancer.

11. Cancer Treatment : Effective cancer therapy through immunomodulation: dynamics of treatment, pharmacogenetics in cancer treatment, cellular senescence in cancer treatment, treating cancer's kinase addiction.

12. p53 pathway : p53 and cancer**,** p53 restoration in tumors**,** p53 pathway: positive and negative feed-back loops**,** Micro RNA and SiRNA in cancer.

Books Recommended:

1. *Fundamental Virology*
   By: Fields and Knipe
   Publishers: Raven
1. *General Virology*

By: Luria
2. *Diagnostic Virology*
   By: G.D. Hseurg
   Publishers: Yale University Press
3. *Biosynthetic Products for Cancer Chemoh*
   By: G.R. Pllil, G.M. Gragg and C.L. Herold
   Publishers: Elseur
4. *Medical Virology*
   By: F.J. Franes & D.O.White
   Publishers: Academic Press
5. *Molecular Biology & Mutagens and Carcinogen*
   By: B. Singer & D. Gran berger
   Publishers: Plenum Press

Paper Code	242913	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Bio-Chemistry of Natural Products**			

1. **Natural Products**: Sources (plants, animals, microbial, marine), classification on chemical basis, role of natural products in development of medicinal chemistry.

2. **Alkaloid:** Sources, classification, isolation, purification & structure determination of nicotine, atropine, morphine, heroine & papaverine, medicinal importance, biosynthesis of some important alkaloids.

3. **Antibiotics:** Source, classification, isolation, structure determination & synthesis of penicillin, semisynthetic penicillin, ampicillin, cephalosporin & tetracycline, mode of action of antibiotics, antibiotic resistance, biosynthesis of some common antibiotics.

4. **Steroids:** Isolation, purification, structure determination & synthesis of some important steroids, cholesterol, ergosterol & stigmasterol. Use of steroids as oral contraceptives.

5. **Terpenoids:** Source, classification, isolation, purification & structure determination of citral, limonene, α-terpeniol, α-pinene, camphor. Medical importance.

6. **Flavonoids:** Source, classification, isolation, purification & structure determination of flavones, flavonols, isoflavones, quercetin. Medical importance of some important flavonoids as antioxidants.

7. **Bioactive compounds from marine sources (Curacin A) & from animal source (epibatidine).**

Books Recommended:

1. *Organic Chemistry,* Vol. 1 and Vol. 2

By: I. L. Finar Publisher: English Language Book
Society and Longman Group. Ltd.
2. *Chemistry of Organic Natural Products,* Vol. 1 and Vol. 2
By: O. P. Agarwal, Publisher: GOEL Publishing House
3. *The Alkaloids*
By: Manske
4. *The Chemistry & Biology of Antibiotics*
By: Vladimis Betina,
Publisher: Elsevier Scientific Publishing Company
5. *Methods in Plant Biochemistry*
By: Harbone

Paper Code	242915	**Marks : 100**	**Credits : 4**	**Class Hours : 60 hrs.**
Paper Title :	**Biostatistics**			

1. Introduction : Research and experimentation, the role of statistics, the nature of statistics, scope of biostatistics.
2. Sampling techniques : Definition, sampling procedure (simple random sampling).
3. Basic statistical principles and terminology: Populations and parameters, samples and statistics, variables, statistical characterization of samples, distributions, statistical concepts pertaining to interpretation and decision.
4. Sample description : Estimation of uncertainty.
(i) Calculation of the mean, variance, and standard deviation
(ii) Machine method of calculating the variance and standard deviation
(iii) Estimation of the standard deviation from the range
(iv) Standard deviation of the mean
(v) Confidence limits of a mean
5. Test of significance :
(i) Measured data : The t test - the t test in paired experiments, the t test in non- paired experiments, selection of the appropriate method of calculating, confidence limits of a difference between means.
(ii) Analysis of variance : The F test - single classification data, single classification data with subgroups, multiple classification data, Duncan's multiple range test, least significant difference test, relationship between t and F, general remarks.
(iii) Enumeration data: The Chi-square (X^2) test - the 1 x n table, the 2 x n table, the use of X^2 with occurrence-nonoccurrence data, X^2 analysis of a 2 X 2 or four-fold table, alternate methods of calculating X^2 , tests of significance when cell frequencies are small, general remarks.
6. Statistical methods : Binomial distribution, Poisson distribution, normal distribution.
(i) Confidence limits of the mean
(ii) Significance of a difference between rates
7. The relationship between variables : Correlation, linear regression.

(i) The "least squares" regression line

(ii) Test for linearity of a regression
(iii) Confidence limits of the regression coefficient
(iv) Significance of a difference between regression coefficients.

8. Dosage-response data :

(i) The estimation of 50 per cent endpoints - graphic approximation of an ED_{50} value, the Reed-Muench method of estimating 50 per cent endpoints.
(ii) The estimation of relative potency.

9. Nonparametric Tests

10. Statistical Package for Social Science (SPSS)

Books Recommended :

1. *An Introduction to Statistics and Probability*
   M.N. Islam
   Book World
2. *Basic Statistics (Methods And Applications)*
   M.A. Jalil and R. Ferdous
3. *Business Statistics*
   S.P. Gupta and M.P. Gupta
   Sultan Chand & Sons.
4. *Applied Statistics (Handbook for Human Settlements Planning)*
   M.A. Quader Miah
   Asian Institute of Technology
5. *Biostatistics : A Foundation For Analysis in the Health Sciences*
   Wayne W. Daniel
   John Wiley & Sons.

Paper Code	242916	**Marks : 100**	**Credits : 4**	**Class Hours : 60 hrs.**
Paper Title :	**Practical**			

1. Isolation of DNA from plant tissue.
2. Isolation of plasmid DNA.
3. Agarose gel electrophoresis of DNA and RNA.
4. Isolation of microbes from natural habitats resistant organisms.
5. Isolation of microbes from natural habitats antibiotic resistant organisms.
6. Determination of biological and chemical oxygen demand (BOD, COD) of different water samples.
7. Determination of bacterial load from different environmental samples.
8. Isolation of lactose fermenting bacteria.
9. SDS polyacrylamide gel electrophoresis of rat liver proteins.
10. TLC of fruit juices.
11. Gel filtration chromatography for separation of known proteins.
12. Estimation of streptomycin.
13. Estimation of serum Vitamin A.
14. Analysis of food stuffs – estimation of moisture, protein, fat, and ash content.

15. Identification of blood groups.
16. Collection of serum and plasma.
17. Total and differential white cell counts.
18. Separation of blood leucocytes.
19. Determination of ESR.

Books Recommended :

1. *Current Protocol of Molecular Biology*
   By: F.M. Ausbel et al**.**
2. *Molecular Cloning*
   By: Sambrok et al.

Paper Code	242918	**Marks: 100**	**Credits: 4**	
Paper Title:	**Viva-voce**			